SPE-227069-MS

A Damage Risk Assessment Method for Crossing Pipelines Based on Non-Contact Magnetic Detection and Coercive Force

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

The crossing pipelines exhibit a relatively high potential risk due to environmental factors. Currently, limited by the confined inspection space and the thick anti-corrosion coating on the pipeline surface, there is no efficient and feasible method for detecting and assessing the damage risks of crossing pipelines. In this study, a strong-weak magnetic composite detection method is proposed to address the inspection challenges. Non-contact magnetic detection technology is utilized for preliminary screening, and the coercive force stress detection technology is employed for positioning and assessing the damage level. Firstly, a force-magnetic coupling magnetization model for steel pipelines is established, and the variation laws of the initial magnetization intensity and coercive force of ferromagnetic materials with the increase of stress under weak magnetic and alternating saturated excitation are studied. Subsequently, a strong-weak magnetic composite detection method and a grading assessment method for damage risks of crossing pipelines are established. Finally, engineering applications are carried out to demonstrate the feasibility and accuracy of this method. The results show that under weak magnetic excitation, the initial magnetization intensity of the pipeline increases with the rise of stress, while under alternating saturated excitation, the coercive force increases exponentially with the increase of stress. In practical applications, 11 damaged areas are identified, including 4 low-risk areas of Level III and 7 medium-risk areas of Level II. Through the structural deformation inspection of the suspension bridge tower and X-ray detection verification, consistent assessment results are confirmed. This proves the feasibility and engineering applicability of the strong-weak magnetic composite detection method, and provides a novel approach for the damage risk assessment of crossing pipelines.

Keywords: Crossing pipelines, Strong-weak magnetic composite detection, Non-contact magnetic detection, Coercive force stress detection

Introduction

Pipeline transportation currently serves as one of the primary methods for oil and gas transmission. In recent years, the increasing demand for oil and gas resources has resulted in extensive construction of long-distance pipelines. Due to their considerable length and large spans, long-distance pipelines are installed in highly complex environments. In special areas such as valleys and river channels, pipelines are often elevated as crossing structures. Suspension crossings represent one of the primary pipeline spanning configurations, utilizing bridge towers and steel cables to support the pipeline. Compared to ground-based pipelines, suspension crossing structures are more vulnerable to damage from natural factors. Influenced by the topographic acceleration effects in mountainous regions, crossing pipeline areas frequently experience intensified wind exposure. Strong winds can induce vortex-induced vibrations between the pipeline and suspension cables, generating cyclic forces that lead to stress concentration and fatigue damage in crossing pipelines. Additionally, the pipeline's self-weight and geological settlement-induced tilting of suspension bridges may cause localized stress exceedance in the pipeline. Severe cases could result in pipeline cracking and safety incidents [1]. Therefore, regular stress detection and safety state evaluation of crossing pipelines are essential to proactively identify potential risk zones.

Current pipeline stress detection technologies are primarily categorized into internal and external inspection methods. Internal inspection methods mainly focus on weak magnetic stress internal detection technology [2,3], alternating current stress measurement (ACSM) [4], and Barkhausen noise detection technology [5]. Liu et al. [2,3] study the correlation between stress and magnetic signals based on a magneto-mechanical coupling model, analyze the influence of external magnetic fields on magneto-mechanical relationships under non-saturated magnetization conditions, and validate the feasibility of weak magnetic internal detection through engineering applications. Duan et al. [4] investigate the distribution patterns of pipeline magnetic fields and eddy current fields using the ACSM method, analyze the effects of probe coil dimensions and structural parameters, and achieve stress concentration zone identification through internal detection experiments. Ma et al. [5] apply Barkhausen noise detection technology to quantitatively evaluate the influence of excitation structure dimensions and intensity on pipeline magnetization fields, determine optimal excitation parameters for continuous Barkhausen noise (CBN) signal acquisition, and demonstrate the feasibility of CBN for long-distance pipeline stress detection. However, internal inspection methods face limitations in crossing pipeline applications due to insufficient propulsion capacity of internal detectors when pipeline pressure is low, particularly at elevated sections of crossing structures. External inspection methods primarily include scanning and point measurement modes. Scanning stress detection technologies encompass laser ultrasonic technology [6,7], ultrasonic phased array technology [8,9], and non-contact magnetic detection technology [10,11,12]. Marusina et al. [6] utilize the UDL-2M laser ultrasonic flaw detector for residual stress detection in steel, confirming the method's feasibility. Zhan et al. [7] establish numerical relationships between laser physical parameters and applied loads through simulations and experimental testing on thin steel plates, verifying the accuracy of laser ultrasonic stress detection. In ultrasonic phased array applications, Walker et al. [8] implement this technology for Ti-6Al-4V sample inspection and validate results using the contour method (CM). Kono et al. [9] achieve stress corrosion crack detection in austenitic stainless steel welds using a single ultrasonic phased array probe. For non-contact magnetic detection, He et al. [10,11] develop a force-magnetic coupling model to elucidate stress-magnetic signal correlations and quantify these relationships through sealed hydrostatic pressure tests. Li et al. [12] further characterize magnetic signal fluctuations in stress concentration zones. While laser ultrasonic and ultrasonic phased array technologies require direct surface contact, their application is constrained by the fixed clamping design of suspension bridge-mounted pipelines. Non-contact magnetic detection enables rapid scanning of crossing pipelines but can only identify stress-concentrated cross-sections, necessitating complementary point measurement methods for precise localization.

Point measurement stress detection methods mainly include strain gauge measurement technology [13], X-ray diffraction technology [14], and coercive force detection technology [15,16]. Strain gauge measurement technology is primarily used for stress variation monitoring at fixed positions and cannot be applied to mobile measurements. X-ray diffraction technology represents a novel stress detection approach. Gan et al. [14] demonstrate the accuracy of X-ray diffraction in detecting stress distributions through multiple experiments, but also highlight its stringent requirements on material surface conditions and microstructural homogeneity. Coercive force detection technology has recently gained significant attention due to its utilization of magnetic hysteresis effects in metallic materials under different stress states. Feng et al. [15] conduct coercive force detection experiments on X80 steel pipelines, establish a coercive force-stress relationship model under biaxial stress conditions, and verify the feasibility of this technology. Yang et al. [16] prove a linear correlation between stress and coercive force through tensile experiments, developing a stress damage evaluation method based on coercive force parameters with validated results. In summary, coercive force detection technology stands as the most suitable point measurement method for localized stress detection in crossing pipelines.

Stress concentration constitutes a critical factor leading to structural failure in crossing pipelines. Current studies have focused on the stress-strain characteristics of elevated crossing pipelines under static loads, wind forces, earthquakes, and soil settlement, demonstrating that axial tensile stress represents the primary cause of failure in such pipelines. However, damage risk assessment for crossing pipelines faces the following challenges:

1. Limited bridge deck space prevents deployment of large-scale detection equipment;
2. Elevated installation complicates internal inspection due to insufficient propulsion force, causing pipeline blockage risks;
3. Extended spanning distances significantly reduce the efficiency of conventional detection methods;
4. Thick anti-corrosion coatings on pipeline surfaces restrict the application of contact-based detection technologies.

To address these challenges, this study proposes a damage risk assessment method combining weak and strong magnetic detection based on the force-magnetic coupling effect in crossing pipeline steel. The methodology employs non-contact magnetic detection technology for preliminary screening and coercive force detection technology for precise localization and grading. Through theoretical analysis, we establish a risk assessment framework for crossing pipelines and validate its feasibility and engineering applicability via practical implementation. The main contributions and innovations of this research are as follows:

1. Establishment of a force-magnetic coupling magnetization model for steel pipelines, theoretically investigating the mechanisms of non-contact magnetic detection and coercive force stress detection, while analyzing the variation trends of magnetization intensity and coercive force with applied stress.
2. Development of a hierarchical detection strategy (screening-localization-quantification) for damaged pipeline regions, formulating a strong-weak magnetic composite detection methodology.
3. Classification of risk levels based on stress-dependent coercive force variations, establishing a damage risk assessment system for crossing pipelines.
4. Successful engineering application on a crossing pipeline, identifying 11 damaged areas (4 Level III low-risk zones and 7 Level II medium-risk zones), with verification of three typical damaged regions confirming the accuracy and practicality of the composite magnetic detection method, thereby providing a novel approach for damage risk assessment in crossing pipelines.

Technical Principle

Non-contact magnetic detection technology

Steel pipelines generate additional magnetic flux density under geomagnetic magnetization. When localized stress concentration or wall thickness reduction due to corrosion occurs, the force-magnetic coupling effect induces measurable changes in magnetic flux density within a specific range above the pipeline. By collecting magnetic flux density signal fluctuations directly above the pipeline using high-precision magnetic sensors, both damage location and severity can be determined, as illustrated in Figure 1. The non-contact magnetic detection technology employs a gradient measurement method to amplify signal anomalies in critical regions. During detection, two horizontally aligned magnetic sensors spaced at pipeline diameter intervals acquire differential magnetic gradient signals. While this technology offers rapid detection speed and effectively identifies pipeline damage trends, its results only reflect the stress state across entire pipeline cross-sections within the anomaly signal range, failing to resolve precise stress distribution within specific cross-sections. Consequently, risk assessments based solely on this method may exhibit significant errors. To address this limitation, our study integrates non-contact magnetic detection for rapid preliminary screening with coercive force stress detection for quantitative localization, enabling accurate risk assessment of crossing pipelines.

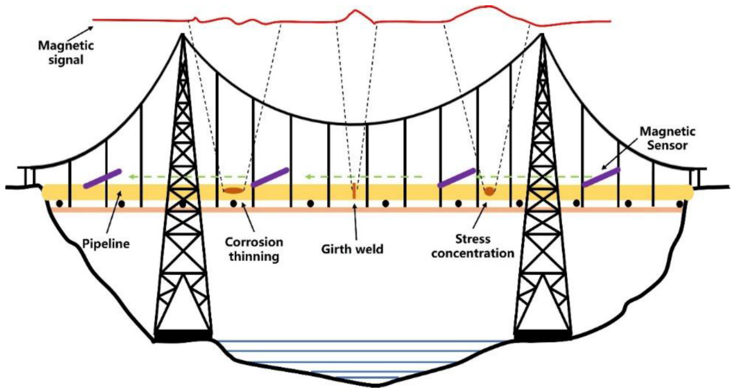


Figure 1—Principle of non-contact magnetic detection technology

Coercive force detection technology

Coercive force refers to the magnitude of reverse magnetic field required to reduce the magnetic flux density of a saturated material to zero, representing the material's resistance to demagnetization. Coercive force stress detection technology was initially proposed by Zakharov in the 1980s [17]. Research indicates that the magnetic domain structure of materials is closely related to their stress state, and alterations in magnetic domain structure directly induce changes in coercive force. Based on the force-magnetic coupling effect, stress measurement can be transformed into coercive force measurement by monitoring coercive force variations to indirectly reflect pipeline stress changes, with the detection principle illustrated in Figure 2. This technology enables both qualitative and quantitative assessment of pipeline stress states, offering rapid detection speed, stable measurement results, minimal surface finish requirements, and no couplant required, demonstrating high engineering applicability.

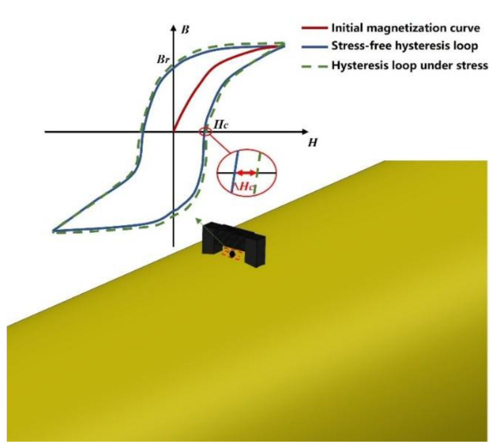


Figure 2—Principles of coercive force stress detection technology

Magnetization model of ferromagnetic materials based on magneto-mechanical coupling effect

The force-magnetic coupling effect describes the influence of stress-strain on the magnetization process of ferromagnetic materials, representing a current research focus in electromagnetic non-destructive testing. Both the non-contact magnetic detection technology and coercive force detection technology proposed in this study achieve pipeline stress detection based on this effect. Building upon the Jiles-Atherton (J-A) model, we establish forward magnetization models under weak and strong magnetic excitation. Through numerical calculations, we analyze stress-induced variations in material magnetic characteristics, theoretically validating the feasibility of stress detection using these two technologies.

In practical detection, magnetic flux density B can be expressed through material magnetization intensity M and external magnetic field intensity H as:

$$B = \mu_0 (H + M) \tag{1}$$

In the equation, μ_0 represents the vacuum permeability. The magnetization intensity M consists of irreversible magnetization M_{irr} and reversible magnetization M_{rev}, whose relationship can be expressed as [18]:

$$\begin{cases} M = M_{irr} + M_{rev} \\ M_{rev} = c\left(M_{an} - M_{irr}\right) \end{cases} \tag{2}$$

In the equation, c represents the reversible coefficient, and M_{an} denotes the anhysteretic magnetization intensity. The Jiles-Atherton (J-A) model employs the Langevin function to simulate the ideal anhysteretic magnetization curve, where the anhysteretic magnetization intensity M_{an} can be expressed as [19]:

$$M_{an} = M_s\left[\coth\left(\frac{H_e}{a}\right) - \frac{a}{H_e}\right] \tag{3}$$

In the equation, M_s represents the saturation magnetization intensity of the material, and a denotes the shape parameter characterizing the hysteresis loop. Under uniaxial stress, H_e represents the effective magnetic field combining the external magnetic field, applied stress, and intrinsic material magnetization, can be expressed as [20]:

$$H_e = H + \alpha M + \frac{3\sigma}{\mu_0}\left[\left(\gamma_1 + \gamma'_1\sigma\right)M_{an} + 2\left(\gamma_2 + \gamma'_2\sigma\right)M_{an}^3\right] \tag{4}$$

In the equation, α represents the domain coupling parameter indicating the interaction strength between magnetic domains, and σ denotes the applied stress. The constants γ, γ'_1, γ_2 and γ'_2 are associated with the magnetostrictive properties of the ferromagnetic material. Based on the principle of energy conservation, the actual magnetization energy equals the anhysteretic magnetization energy minus the hysteresis loss energy, expressed as [21]:

$$M = M_{an} - \frac{k_{eff}\delta}{\mu_0}\frac{dM_{irr}}{dH_e} \tag{5}$$

In the equations, M_{an} denotes the anhysteretic magnetization intensity, k_{eff} represents the pinning coefficient of the material, and δ is the directional coefficient. When $\frac{dH}{dt} > 0,\ \delta = 1$; when $\frac{dH}{dt} < 0,\ \delta = -1$. By coupling Equations (1)–(5), the differential relationship between magnetic flux density B and external magnetic field H can be expressed as:

$$\frac{dB}{dH} = \mu_0\left\{1 + \frac{-(1-c)\frac{\mu_0(M_{an}+H)-B}{k_{eff}\delta} - c\frac{dM_{an}}{dH}}{(1-c)\frac{\mu_0(M_{an}+H)-B}{k_{eff}\delta}\left\{\alpha + \frac{3\sigma}{\mu_0}\left[\left(\gamma_1 + \gamma'_1\sigma\right) + 6\left(\gamma_2 + \gamma'_2\sigma\right)M_{an}^2\right]\right\} - 1}\right\} \tag{6}$$

The derivative of the anhysteretic magnetization intensity M_{an} with respect to the external magnetic field H is expressed as:

$$\frac{dM_{an}}{dH}=\frac{M_s\left[-\operatorname{csch}^2\left(\frac{H_e}{a}\right)\frac{1}{a}+\frac{a}{H_e^2}\right]}{1+M_s\left[\operatorname{csch}^2\left(\frac{H_e}{a}\right)\frac{1}{a}-\frac{a}{H_e^2}\right]\left\{\alpha+\frac{3\sigma}{\mu_0}\left[\left(\gamma_1+\gamma_1'\sigma\right)+6\left(\gamma_2+\gamma_2'\sigma\right)M_{an}^2\right]\right\}} \tag{7}$$

This study employs the ordinary differential equation solver function ode45 in MATLAB software for numerical solution and analysis of the theoretical model, with input parameters determined based on the material properties of X80 steel as follows:

$M_s = 1.585\times10^6$ A/m, $\mu_0 = 4\pi\times10^{-7}\ T\cdot m^2/\mathrm{A}$, $\alpha = 0.001$, $a = 1000$ A/m, $k_{eff} = \frac{1}{270}$, $c = 0.25$, $\gamma_1 = 7\times10^{-18}\ \mathrm{A}^{-2}\cdot\mathrm{m}^2$, $\gamma_1' = -1\times10^{-25}\ \mathrm{A}^{-2}\cdot\mathrm{m}^2/\mathrm{Pa}$, $\gamma_2 = -3.3\times10^{-30}\ \mathrm{A}^{-2}\cdot\mathrm{m}^2$, $\gamma_2' = 2.1\times10^{-38}\ \mathrm{A}^{-2}\cdot\mathrm{m}^2/\mathrm{Pa}$. The external magnetic field intensity is set within the range of 0~20 kA/m to analyze the variation trend of magnetic flux density with increasing magnetic field intensity during the initial magnetization phase. The resulting initial magnetization curve is plotted as shown in Fig. 3.

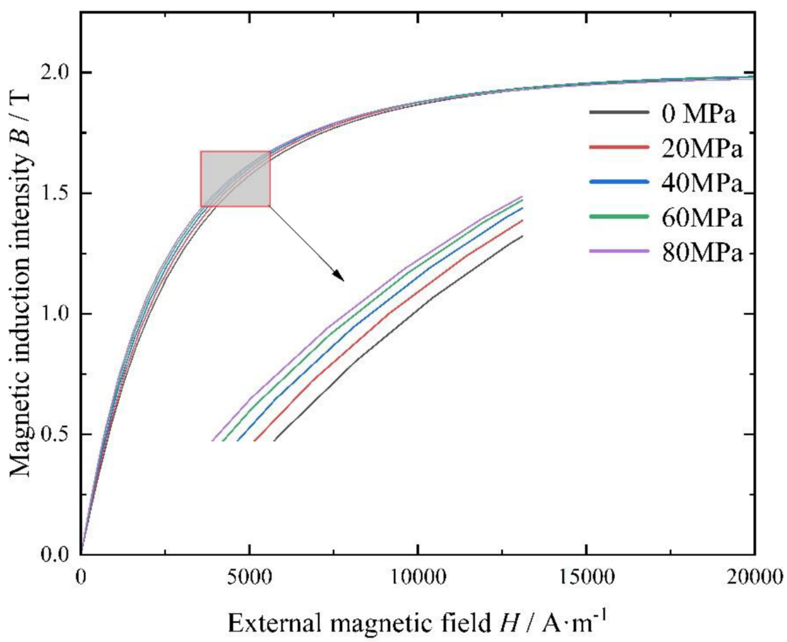
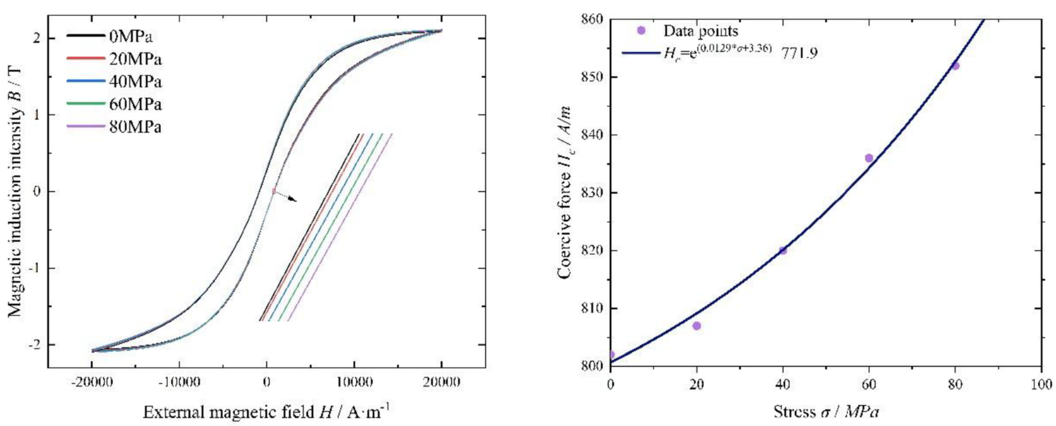


Figure 3—Initial magnetization curve

As illustrated in Figure 3, during the initial magnetization phase, the material's magnetic flux density increases rapidly with the enhancement of external magnetic field intensity, but the growth rate gradually decreases. When the external magnetic field intensity reaches a sufficiently high level, the magnetic flux density approaches saturation, indicating the material has attained its saturation magnetic flux density. Under additional stress, the trend of the magnetization curve remains unchanged, but the growth rate of magnetic flux density varies. At identical external magnetic field intensities below saturation, higher additional stress results in greater magnetic flux density and larger variation amplitudes. Once the magnetic flux density saturates, the saturation value remains consistent across different stress conditions. Non-contact magnetic detection technology exploits this magnetic characteristic of steel pipelines. By utilizing the weak geomagnetic field to induce unsaturated magnetization in pipelines, it measures magnetic flux density variations across different pipeline locations. Abnormal fluctuations in magnetic flux density within specific regions indicate localized stress concentration, thereby enabling rapid screening of stress-induced damage.

The external magnetic field intensity is cycled between 20 kA/m → -20 kA/m → 20 kA/m to simulate alternating saturation excitation magnetization of the material within a short duration. Solving the differential equation derived from Formula (6) yields a complete hysteresis loop, as shown in Fig. 4(a). Observations indicate that under applied stress, the hysteresis loop exhibits a "widening" trend: greater applied stress correlates with increased loop width and amplified widening magnitude. At stress levels of 0 MPa, 20 MPa, 40 MPa, 60 MPa, and 80 MPa, the material's coercive force measures 802 A/m, 807 A/m, 820 A/m, 836 A/m, and 852 A/m, respectively. The stress-dependent variation trend of coercive force is plotted in Fig. 4(b), demonstrating an exponential increase with rising stress. By applying strong alternating magnetic excitation to steel pipelines and measuring coercive force, the coercive force stress detection technology accurately reflects stress distribution and enables precise assessment of pipeline stress damage risk levels.



(a) (b)

Figure 4—Change law of hysteresis loop under different stresses and trend of coercive force: (a) Hysteresis loop; (b) Stress-coercive force change curve

Method for assessing the risk level of stress damage across pipelines

Detection Instruments

The non-contact magnetic detection for crossing pipelines employs the triaxial high-precision pipeline magnetic gradient signal detection device independently developed by the research team, which accurately captures the three-axis magnetic gradient components and their modulus of steel pipelines. The magnetic stress detection system consists of a non-contact magnetic gradiometer, a data acquisition and transmission unit, an engineering host computer, and magnetic stress detection software, as shown in Fig. 5. Key specifications include a detection resolution of 0.1 nT/m, noise level ≤30 pT/√Hz @1 Hz, and an operational temperature range of -40°C to +80°C.

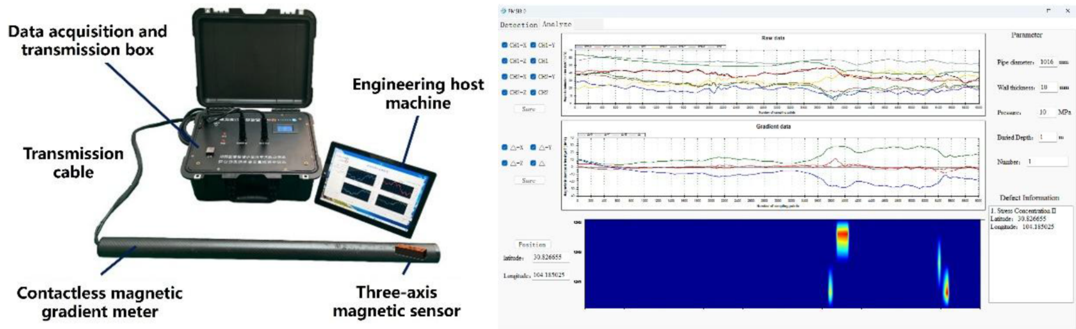


Figure 5—Three-axis high-precision pipeline magnetic gradient signal detection system:(a) Non-contact magnetic detection device; (b) Detection software

The coercive force stress detection system adopts a mature commercial coercive force tester, as shown in Fig. 6. Its detection probe consists of a high-frequency electromagnetic excitation device and magnetic sensors. By applying high-frequency alternating electromagnetic excitation to the pipeline and recording the residual magnetic field intensity under zero excitation conditions, the system accurately measures the pipeline's coercive force. Key specifications include a maximum excitation field of 20,000 A/m and an operational temperature range of -20°C to +50°C. Prior to use, the device must be calibrated using a zero-stress test block made of the same material as the pipeline to determine the material's baseline coercive force, thereby establishing a foundation for quantitative stress analysis.

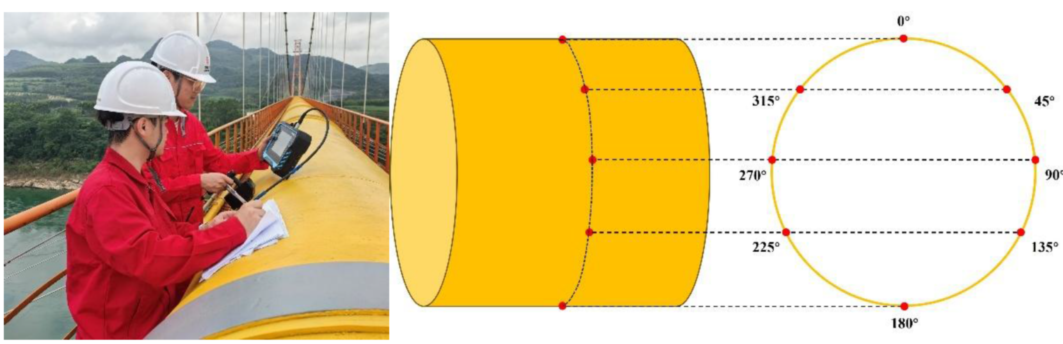
Figure 6—Coercive force tester

Detection method

The on-site inspection of crossing pipelines follows a non-contact magnetic preliminary screening → coercive force stress detection → quantitative localization strategy. During field operations, the operator manually moves the magnetic gradiometer along the marked pipeline route at a constant speed. The gradiometer is kept horizontal, positioned directly above the pipeline and perpendicular to it, with a detection height of 0.2 m, as shown in Fig. 7. Throughout the inspection, the gradiometer automatically collects magnetic field data from the ferromagnetic pipeline and transmits it to the host computer software. When damage or defects exist in the pipeline, the magnetic gradient signals exhibit abnormal fluctuations. Pipeline damage locations are identified by detecting peaks in the magnetic gradient modulus, while performing multiple scans on magnetic anomaly areas to ensure detection repeatability and accuracy. Confirmed damage zones are then marked. After completing the non-contact magnetic preliminary screening across the entire pipeline, coercive force detection is conducted on the identified damage areas.

Figure 7—The initial screening detection method for non-contact magnetic detection pipeline damage detection

When inspecting in-service crossing pipelines under pressurized conditions, the proposed method requires pressure-included calibration in addition to zero-stress calibration to establish the baseline coercive force of the pipeline under internal pressure. After completing non-contact magnetic preliminary screening, multiple anomaly-free areas along the pipeline are selected for coercive force detection to evaluate the normal coercive force level under operational pressure. Once calibrated, coercive force detection is performed on the marked magnetic anomaly areas. Eight evenly distributed directional points are measured around the pipeline cross-section in magnetic anomaly regions, as shown in Fig. 8. During testing, the electromagnetic excitation core must maintain tight contact with the pipeline surface, with repeated measurements at the same location to ensure accuracy.



(a) (b)

Figure 8—Coercive force detection method:(a) Coercive force detection; (b) Coercive force circumferential detection orientation

Evaluation method of damage risk level

To accurately assess the damage severity in magnetic anomaly areas of crossing pipelines, this study introduces the coercive force-stress safety factor $H_{c\sigma}$ for crossing pipelines, based on the coercive force under zero-stress conditions H_{c0} and the coercive force under internal pressure H_{cp}. The safety factor is defined as:

$$H_{c\sigma} = H_{cp} - H_{c0} \tag{8}$$

The study reveals an intrinsic coercive force growth trend in pipelines with identical operating conditions and material composition during in-service operation. Based on a multiple factor of $H_{c\sigma}$, the stress-induced damage severity in crossing pipelines is classified into three risk levels: low, medium, and high. The risk classification methodology is detailed in Table 1.

Table 1—Methods for dividing damage risk levels

Coercive force range	Level classification	Damage Status	Treatment measures
$H_c > H_{cp} > 5H_{c\sigma}$	I	High risk	Fix it now
$H_{cp}+2H_{c\sigma} < H_c \# H_{cp}+5H_{c\sigma}$	II	Medium risk	Planned fixes
$H_{cp} < H_c \# H_{cp}+5H_{c\sigma}$	III	Low risk	Regular detection

Engineering Application

Pipeline Overview

The engineering application is implemented on a crossing pipeline section of the China-Myanmar Natural Gas Pipeline, as shown in Fig. 9. The crossing adopts a suspension bridge structure with a main span of 300 m and a horizontal pipeline length of 392 m. The natural gas pipeline at the crossing has a design pressure of 10 MPa and uses X80 longitudinal submerged arc welded (LSAW) steel pipes with dimensions of Φ1016 × 22.9 mm.

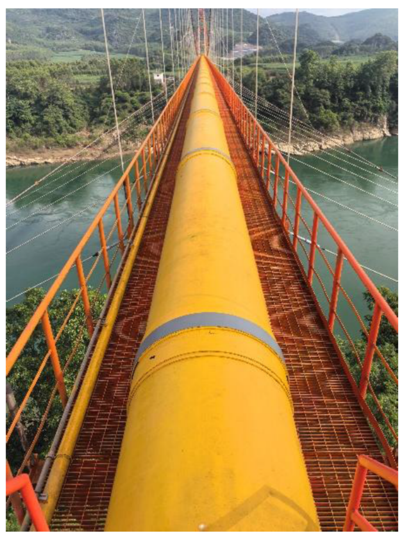


Figure 9—A certain crossing pipe section of the China-Myanmar trunk gas pipeline

Zero-stress calibration test blocks are fabricated using the pipeline material and calibrated with the coercive force tester. The coercive force under zero-stress conditions H_{c0} is 5.1 A/cm, while the operational coercive force under internal pressure H_{cp} is 5.7 A/cm, yielding a coercive force-stress safety factor $H_{c\sigma}$ of 0.6 A/cm. The damage risk classification criteria are provided in Table 2.

Table 2 Risk level of damage across segments

Table 2—Risk level of damage across segments

Coercive force range	Level classification
$H_c > 8.7$	I
$6.9 < H_c \# 8.7$	II
$5.7 < H_c \# 6.9$	III

Detection results

Non-contact magnetic preliminary screening is first conducted on the crossing pipeline, with the detection results shown in Fig. 10. The screening detects 11 magnetic anomaly regions, including 2 pipe body anomalies located at both ends of the crossing section and 9 girth weld anomalies concentrated in the central portion. Circumferential coercive force detection is performed on these 11 anomaly regions. The maximum coercive force values and corresponding damage risk classifications for each anomaly are summarized in Table 3, indicating 7 regions with Level II risk and 4 regions with Level III risk.

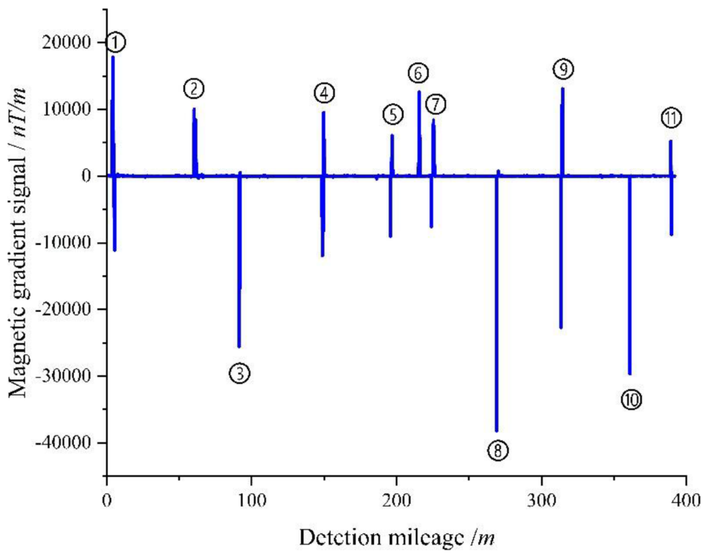
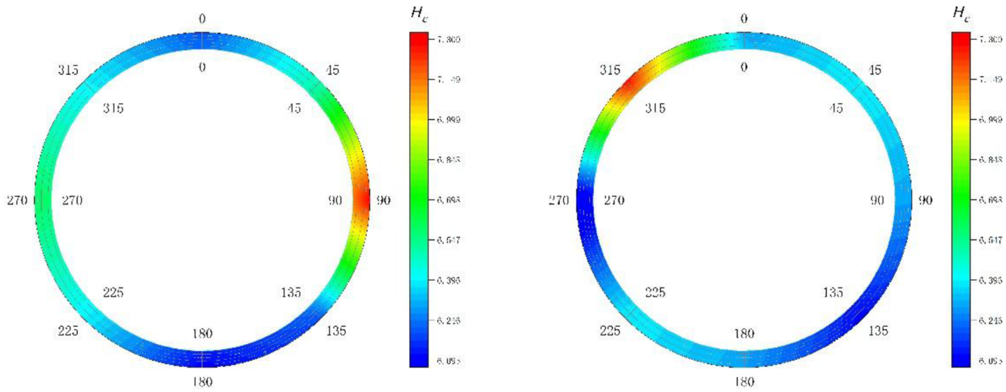


Figure 10—The initial screening results of non-contact magnetic detection

Table 3—Coercive force detection results in abnormal areas

Serial number	Maximum coercive force	Risk level	Damage Status
1	7.3	II	Medium risk
2	6.6	III	Low risk
3	7.7	II	Medium risk
4	7.6	II	Medium risk
5	6.5	III	Low risk
6	6.8	III	Low risk
7	6.9	III	Low risk
8	8.1	II	Medium risk
9	7.4	II	Medium risk
10	7.7	II	Medium risk
11	7.3	II	Medium risk

Two pipe body magnetic anomaly regions (No. 1 and No. 11) and the anomaly region with the highest magnetic amplitude (No. 8) are selected for analysis. The circumferential coercive force detection results for regions No. 1 and No. 11 are shown in Fig. 11. The coercive force of region No. 1 reaches its maximum value of 7.3 A/cm at the 90° direction, while region No. 11 achieves the same maximum value (7.3 A/cm) at the 315° direction. Both pipe body anomalies exhibit identical magnitudes and symmetrical positions without visible bending deformation. This pattern is analyzed to correlate with the support conditions of the suspension bridge structure.



(a) (b)

Figure 11—Circumferential coercive force detection results for the abnormal magnetic area of the tube body: (a) Coercivity heat map for area No.1; (b) Coercivity heat map for area No.2

A review of the suspension bridge structural inspection results reveals minor angular offsets and tilts caused by geological settlement. Specifically, the North Bank Tower (near the No.1 anomaly) exhibits a left-side longitudinal displacement of 6.3 cm southward, a right-side longitudinal displacement of 7.4 cm southward, and a lateral displacement of 6.7 cm leftward. The South Bank Tower (near the No.11 anomaly) shows a left-side longitudinal displacement of 0.6 cm northward, a right-side longitudinal displacement of 0.4 cm northward, and a lateral displacement of 5.8 cm leftward. These tower displacements induce slight bending stress on both ends of the crossing pipeline, macroscopically manifesting as localized stress concentration in the pipe body, with a risk level aligning with the moderate-risk planned maintenance strategy.

The No.8 anomaly region is located at the central girth weld of the crossing pipeline. Its circumferential coercive force detection results are shown in Fig. 12. The coercive force reaches a maximum value of 8.1 A/cm at the 225° direction and a minimum of 7.0 A/cm at the 180° direction. With the circumferential stress concentration across the entire girth weld being notably high, the observed stress concentration is attributed to the presence of weld defects.

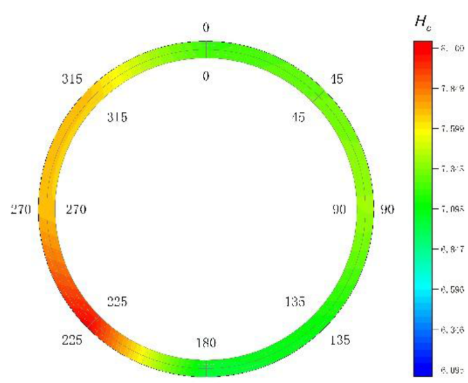
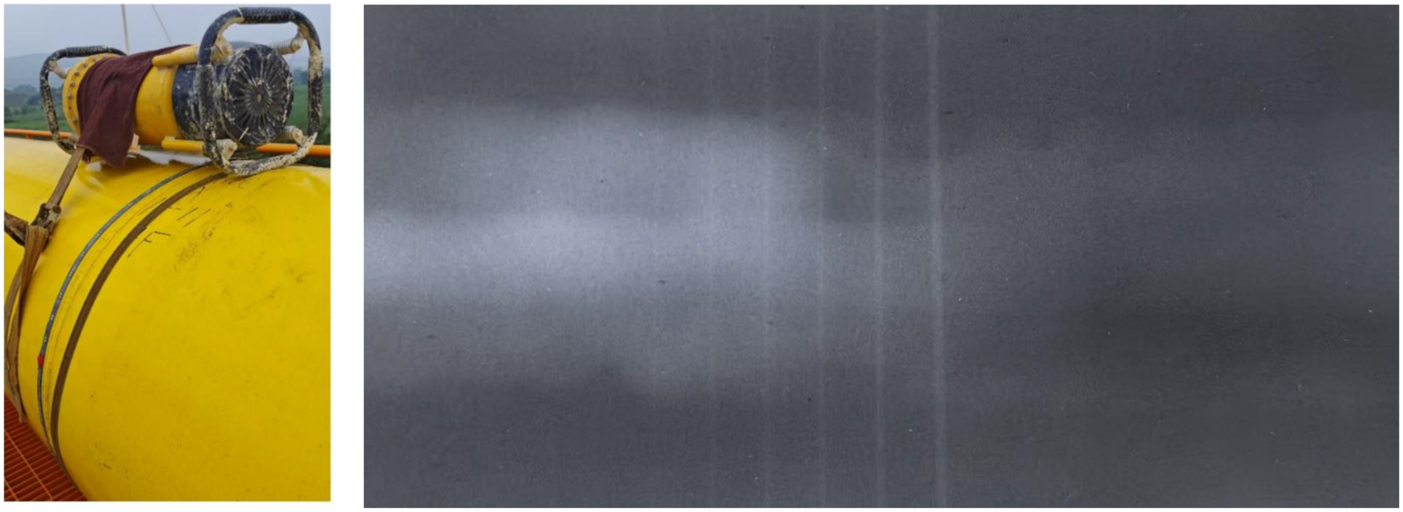


Figure 12—Circumferential coercive force detection results for abnormal areas No. 8

To determine the overall damage state of the No.8 girth weld region, X-ray testing is conducted as shown in Fig. 13. The results reveal seven spherical porosity indications and an internal undercut defect of 11 mm near the 225° direction. According to the SY/T 4109-2020 "Non-destructive Testing of Petroleum and Natural Gas Steel Pipelines" standard, the weld is rated as Level II, requiring scheduled repair. This conclusion aligns with the risk classification assessment results, further validating the effectiveness and engineering applicability of the combined non-contact magnetic detection and coercive force stress detection method for damage risk evaluation in crossing pipelines.



(a) (b)

Figure 13—X-ray field detection of ring welds in abnormal area No. 8: (a) on-site inspection; (b) radiographic testing results

Conclusions

This study addresses the current challenges in damage risk assessment for crossing pipelines by proposing a strong-weak magnetic composite detection method that combines non-contact magnetic screening for preliminary inspection with coercive force stress detection for localization and classification. Theoretical analysis clarifies the mechanism of non-contact magnetic detection and the stress-dependent variation of coercive force, establishing a damage risk assessment methodology validated through engineering applications. Key conclusions include:

1. Under weak magnetic excitation, the initial magnetization intensity of ferromagnetic materials positively correlates with stress, while saturation magnetization remains unaffected.
2. Coercive force increases exponentially with stress under alternating saturation excitation, serving as a reliable indicator of stress concentration.
3. The proposed composite detection method proves feasible and efficient for crossing pipeline inspection, enabling effective screening of stress-concentrated areas.
4. The damage risk classification framework accurately quantifies stress-induced damage severity, providing a novel strategy for pipeline safety monitoring.

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